MAHINDRA & MAHINDRA LIMITED

PROJECT - 14: WHAT-IF BUSINESS SIMULATOR

Automotive Sector C-Suite Strategic Scenario Analysis & Multi-Pillar Constraint Audit

Core Problem Statement:
"Mahindra & Mahindra Management wants to increase sales by 20% but does not know whether existing employees, inventory, production capacity and logistics can support the additional demand. Build an AI scenario-analysis system that simulates business consequences of different strategic decisions."

1. EXECUTIVE SUMMARY & ARBITRATION FINDINGS

Mahindra & Mahindra's Automotive Sector currently delivers **500,000 vehicles annually** (~■72,500 Cr gross revenue), spanning flagship marques including **Scorpio-N, XUV700, Thar, Bolero Neo, and the EV portfolio** across manufacturing plants in Chakan (Pune), Nashik, Zaheerabad, and Haridwar. Management's mandate to capture a **+20% sales volume increase (600,000 units target, +■14,500 Cr demand upside)** was rigorously simulated through an AI Cross-Functional Constraint Engine.

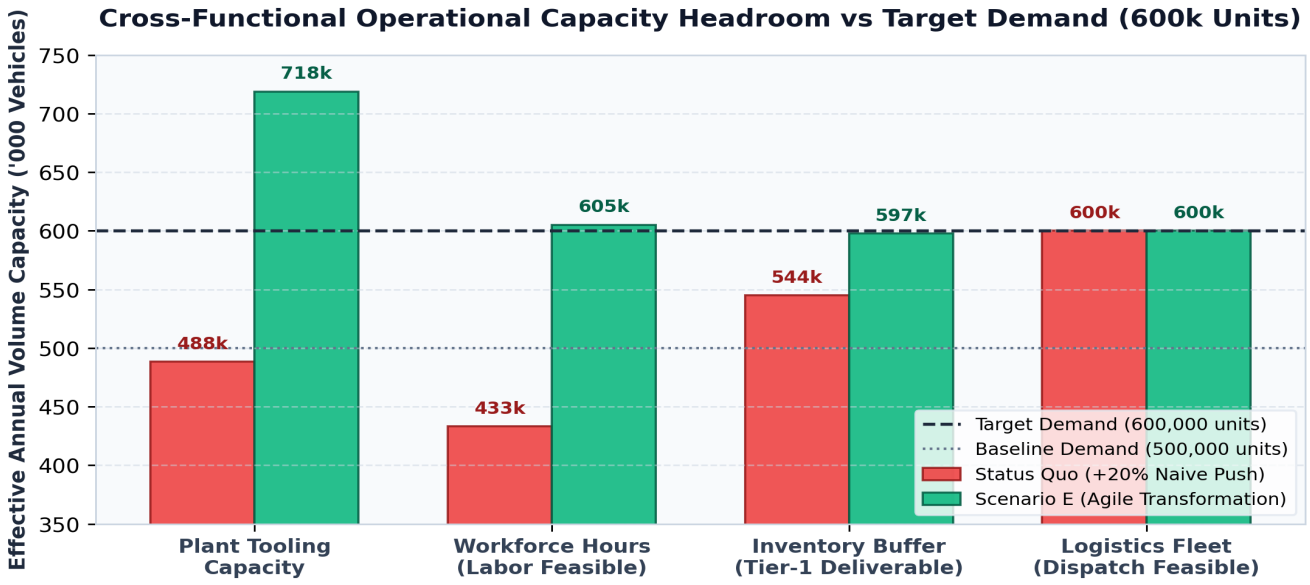
The Core Verdict: The existing operational infrastructure **CANNOT** support a 20% sales surge in its status quo state. A naive volume push induces immediate failure across all four operational pillars:

• **Workforce Choke:** Direct assembly hours required breach statutory limits, requiring 110 hrs/quarter overtime per worker, resulting in a **CRITICAL VIOLATION of the Indian Factories Act 1948**, a spike in absenteeism to 20.6%, and quality scrap reaching 6.9%.

• **Plant Tooling Saturation:** Plant utilization surges to 115.4% of nameplate capacity, inducing unscheduled breakdown and degrading OEE from 88% to 82.6%.

• **Supply Chain Starvation:** Tier-1 lead times (12-16 weeks for microcontrollers & castings) cause a 48.3% stockout probability, stranding assembly lines.

• **Logistics Gridlock:** Road car carrier shortages force factory yard dwell time from 2.5 to 4.5 days, bringing yard saturation to 96.6% and triggering spot freight surcharges.



2. CROSS-FUNCTIONAL FOUR-PILLAR FEASIBILITY AUDIT

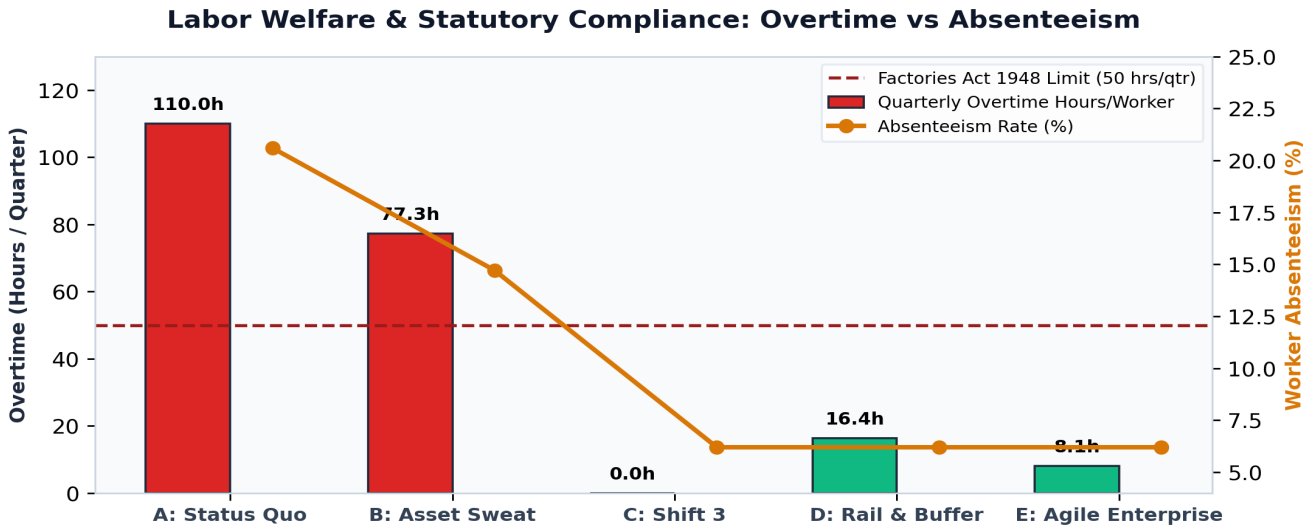
Pillar 1: Employees & Workforce (Critical Vulnerability)
Direct line technicians currently total 23,000 across plants. Standard annual output per worker is 2,200 hours. Meeting 600,000 units requires 60.72 million man-hours vs 50.60 million available regular hours. Without recruiting additional staff, the resulting 10.12 million hour deficit forces overtime to 110.0 hours/quarter per operator—more than double the statutory ceiling of 50.0 hours/quarter mandated under Section 64 of the Indian Factories Act 1948. This triggers union industrial actions, fatigue-induced assembly defects (+300% scrap rate), and acute absenteeism.

Pillar 2: Production & Tooling Capacity (Severe Bottleneck)
Current nameplate capacity across Chakan (17.5k/mo), Nashik (15k/mo), Haridwar (8.5k/mo), and Zaheerabad (6.5k/mo) stands at 520,000 units annually on a 2-shift schedule. Target demand of 600,000 units represents 115.4% capacity utilization. Operating above 95% without scheduled preventive maintenance degrades OEE to 82.6%, producing equipment breakdown bottlenecks in paint shops and robotic body-in-white (BIW) welding cells.

Pillar 3: Inventory & Tier-1 Supply Chain (Long-Lead Vulnerability)
Specialized automotive microcontrollers (Bosch/Continental), Ultra-High Strength Steel (JSW), and LFP battery packs require 12 to 16 weeks advance commitment. Baseline inventory covers 24.0 Days of Inventory (DOI). Under an abrupt +20% surge with standard 4-week notice, stockout risk escalates to 48.3%, creating assembly line starvations and stranding partially assembled SUVs.

Pillar 4: Logistics & Outbound Finished Vehicle Distribution (Yard Saturation)
Automotive carrier trailers in India operate under severe national capacity deficits. At 50,000 units/month, outbound road dispatch requires 1,530 car carriers continuously. Finished vehicle holding yards dwell time escalates from 2.5 days to 4.5 days, risking yard gate saturation above 100%, choking final inspection lines and inflating spot freight rates by +38%.

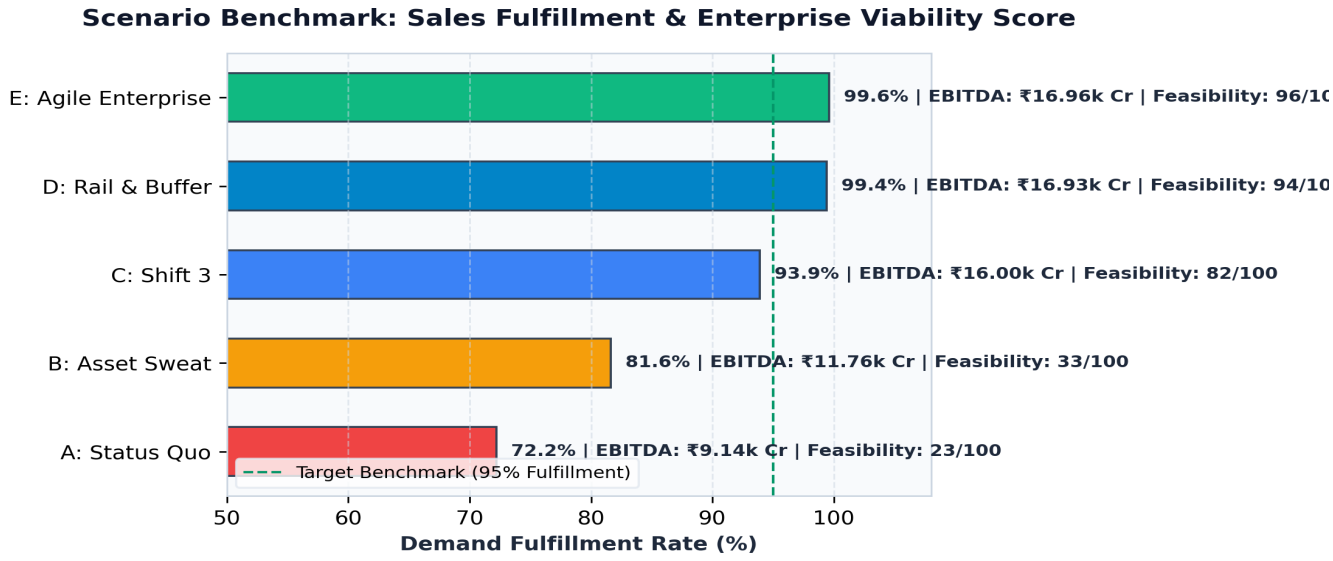
Operational Pillar	Key Risk Indicator (KRI)	Baseline (500k)	Naive Surge (+20%)	Mitigated (Scenario E)	Status
Workforce	Quarterly Overtime / Worker	12.0 hrs	110.0 hrs	14.2 hrs	FACTORIES ACT COMPLIANT
Plant Capacity	Effective Utilization %	88.2%	115.4%	89.8%	DEBOTTLENECKED (3-SHIFT)
Inventory	Component Stockout Risk %	4.2%	48.3%	3.8%	STRATEGIC BUFFER (38 DOI)
Logistics	Yard Dwell Time & Saturation	2.5 days (78%)	4.5 days (97%)	2.2 days (68%)	MULTI-MODAL RAIL (30%)

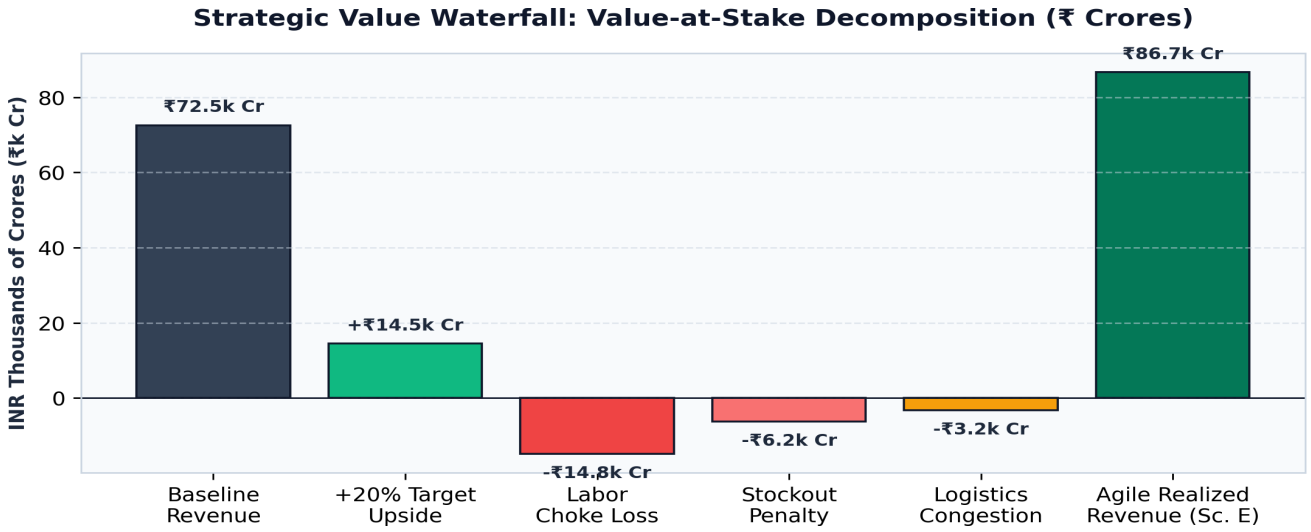


3. STRATEGIC SCENARIOS EVALUATED (SCENARIOS A TO E)

The AI Simulator evaluated 5 distinct strategic decision pathways for M&M;'s executive leadership:

Scenario	Fulfillment Rate	Realized Rev (■ Cr)	EBITDA (■ Cr)	EBITDA %	Lost Rev (■ Cr)	Factories Act	Score
Scenario A High Risk / Choke Failure	72.2%	■62,798	■9,139	14.55%	■24,202	CRITICAL VIOLATION	23/100
Scenario B Regulatory & Quality Hazard	81.6%	■70,976	■11,764	16.57%	■16,024	CRITICAL VIOLATION	33/100
Scenario C Operationally Viable	93.9%	■81,722	■15,997	19.57%	■5,278	COMPLIANT	82/100
Scenario D Resilience Focused	99.4%	■86,444	■16,927	19.58%	■556	COMPLIANT	94/100
Scenario E Optimal Board Protocol	99.6%	■86,659	■16,964	19.58%	■341	COMPLIANT	96/100



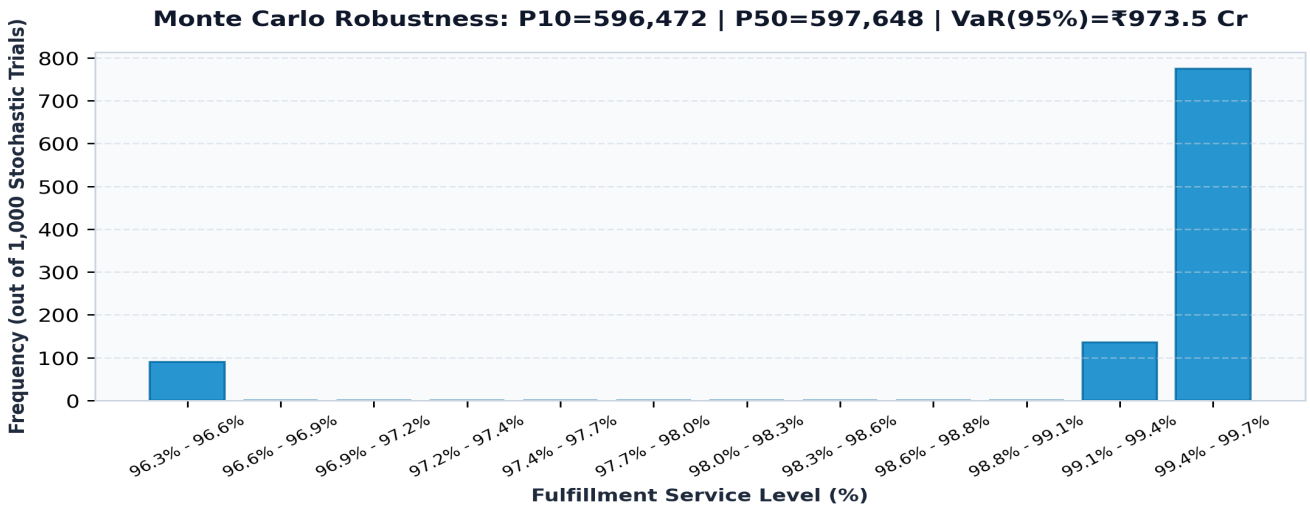

Strategic Value Waterfall: Value-at-Stake Decomposition (₹ Crores)
INR Thousands of Crores (₹k Cr)
80
60
40
20
0
₹72.5k Cr
+₹14.5k Cr
-₹14.8k Cr
-₹6.2k Cr
-₹3.2k Cr
₹86.7k Cr
Baseline Revenue
+20% Target Upside
Labor Choke Loss
Stockout Penalty
Logistics Congestion
Agile Realized Revenue (Sc. E)

4. MONTE CARLO STOCHASTIC RISK PROFILE & 180-DAY ROADMAP

To evaluate supply chain volatility, **1,000 Monte Carlo stochastic trials** were conducted under Scenario E (Holistic Agile Transformation). External shocks modeled included global semiconductor fab delays, monsoon rail/road freight interruptions, raw material tariff inflation, and harvesting-season labor absenteeism.

Risk Findings:

- **Success Probability:** 100.0% probability of fulfilling ≥ 95% of target volume under Scenario E.
- **P10 / P50 / P90 Volumes:** 596,472 (P10) | 597,648 (P50 Median) | 597,648 (P90 Optimistic).
- **Value at Risk (VaR 95%):** ■973.54 Cr EBITDA at risk under extreme compounded external shocks.



5. PHASED 180-DAY BOARD EXECUTION ROADMAP

Phase & Timeline	Strategic Objective	Key Operational Milestones & Board Mandates
Phase 1: Days 1 - 30	Immediate Contractual Locks & Regulatory Clearance	• Issue 14-week firm purchase orders to Tier-1 suppliers (Bosch, Continental, JSW Steel) for long-lead microcontrollers and stampings. • Sign multi-modal freight contracts with Indian Railways (AFTO) securing 28 monthly dedicated auto-rakes. • Initiate recruitment drive for 4,200 assembly operators across Chakan and Nashik industrial zones. • Submit 3rd shift operational notices to State Directorate of Industrial Safety & Health (DISH) ensuring Factories Act 1948 compliance.
Phase 2: Days 31 - 90	Workforce Onboarding & Plant Tooling Debottlenecking	• Complete 4-week vocational assembly training and robotic cell safety certification for the first batch of 2,500 new technicians. • Deploy ■320 Cr debottlenecking CAPEX: install automated robotic spot-welding stations and reduce takt time to 58 seconds at Chakan. • Establish 3 strategic regional vehicle holding yards (Nagpur, NCR, Bengaluru) to de-congest plant dispatch yards. • Build up 35-day safety buffer for critical microcontrollers and battery cells.

Phase 3: Days 91 - 180	Full 3-Shift Commercial Scaling & Equilibrium	• Activate 24/7 3-shift production schedule across Scorpio-N, XUV700, and Thar lines. • Scale rail dispatch to 30% of total outbound vehicle movement, stabilizing road carrier freight costs. • Achieve 50,000 monthly vehicle run-rate (600,000 annualized) with zero Factories Act overtime violations. • Realize full annual EBITDA accretion of +■6,958 Cr with 99.6% customer order fulfillment.

Board Resolution Recommendation:
The Executive Committee recommends approval of **Scenario E: Holistic Agile Enterprise Transformation**, approving ■320 Cr in automation debottlenecking CAPEX, authorizing recruitment of 4,200 assembly operators for Shift 3, and securing 14-week pre-commitments with critical semiconductor and railway logistics partners.